Blssed G/Michael Catholic Secondary School

1st Semester Chemistry test Two for Grade 10th

Dec, 2018 E.C

Time Allowed 40'

Name: Bizunya Wondimagegnehu Sec: 10mB R.No. 12

I. Choose the best answer among the given alternatives.

C 1. All of the following Salt are soluble in water except

A. Na_2CO_3 B. $AgNO_3$ C. $NH_4(NO_3)$ D. $Ca_3(PO_4)_2$

A 2. Which of the following is an example of a solid solution? cu+

A. Copper dissolved in gold. C. Ethanol dissolved in water

B. Glucose dissolved in water. D. sodium chloride dissolved in water

B 3. Which of the following compound can dissolve in water? P

A. Carbon dioxide (CO_2 N.P) B. Ammonia. C. Nitrogen. N_2 (N.P) D. hydrogen H_2 (N.P)

B 4. One of the following is not a property of solution?

A. Is homogenous mixture. C. Its particle size less than 1nm

B. Can filter material out of solution. D does not scatter light

D 5. Which one of the following substances has dispersion (N.P) inter molecular force (N.P)?

(no hydrogen bond, no dipole dipole bond)

A. CH_3OH (P) B. NH_3 (P) C. H_2S (P) D. CH_4

A 6. What is meant by the rate of dissolution of solute?

A. The amount of solute that goes into solution per unit time

B. The ability of solute to go into solution

C. the maximum amount of solute that a pure solvent can hold in solution

D. The dynamic equilibrium attained between solute and solvent

C 7. Which of the following ionic compound have grater lattice energy?

A. NaCl B. CsCl C. KCl D. HCl

C 8. Which of the following intermolecular force highest in strength? 1, Pon-dipole 2, Hydrogen 3. di-dipole

A .London force C. H- bonding

B. Dipole-dipole force D. dipole- induced dipole force

C 9. When 20 g of $CaCO_3$ was dissolved in 80 g of water, the temperature of water rose by $8c^o$; what is the molar heat of the solution in KJ/mole. Take the specific heat capacity of water $4.18J/g.c^o$

A. -33.44kJ/mole B. 16.72kJ/mole C. -16.72kJ/mole D. 33.44kJ/mole

A 10. What kinds of solution forms when water evaporates in air.

A. Gas in gas solution C. liquid in liquid solution

B. gas in liquid solution D. liquid in solid solution

C 11. Which of the following is exothermic in solution formation?

A. separation of solute particle (endo)
B. separation of solute particle (endo)
C. interaction of solvent among solute particle
D. break up of solute among solvent particles

II. Write true if the statement is correct and false if the statement is wrong.

false 12. Both ends of soap molecules are nonpolar.

True 13. The solvent of liquid solution always water.

True 14. A saturated solution cannot dissolve more solute at a given temperature.

True 15. The solubility of ionic substances usually increased with increasing temperature.

false 16. Ionic solid with relatively large lattice energy is usually soluble.

III. Match the intermolecular forces under column A with their examples under column B.

A.	B
B 17. London force. N.P & N.P	A. O_2 in water (N.P) (N.P)
D 18. Dipole dipole force. P & P	B. I_2 in CCl_4 (N.P) (N.P)
A 19. Dipole inducted dipole force. NP & P	C. KCl in water (P) (P)
C 20. Ion dipole force. I & P	D. NH_3 and water (P) (P)
E 21. Ion induced dipole force. I & NP	E. NO_3^- and I_2 (I) (N.P)
	F .HCl and HBr (P) (P)

IV. Fill in the blank spaces with the appropriate answers.

22. A dissolution process that release heat energy to the surrounding is known as Exothermic
23. The process through which a solute particle dissolved in solvent particle to produce solution is dissolution
24. The instrument used to measure enthalpy(heat) of solution in the laboratory known as _____.
25. The process of attraction & association of molecules of solute and solvents is Solvation.

Chemistry Answer Sheet for Mid-2,G. 10 **Time allotted: 40'**

Name Bitanya Wondimagegnehu **Sec.** 10th B **No.** 12

1. D	11. C	21. E
2. A	12. false	22 Exothermic
3. B	13. True	23 Dissolution
4. B	14. True	24. Thermometer
5. D	15. True	25. Solvation
6. A	16. false	
7. A	17. B	
8. C	18. D	
9. C	19. A	
10. A	20. C	

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17 + 2 / 20

Blessed G/Michael Catholic Secondary School

1st Semester Chemistry Test Two for Grade 10th

Nov, 2018 E.C Time Allowed 35'

Name: Bitanya Wondimagegnehu Sec: 10th B R.No. 12

I. Choose the best answer among the given alternatives.

C 1. Which of the following is a heterogeneous mixture?

A . Sugar dissolved in water. C. mixture of oil and water.

B. Salt dissolved in water. D. mixture of alcohol and water

B 2 . Shaving cream produce foam what kind of colloid is shaving cream?

A. Liquid dispersed in gas. C . Solid dispersed in a liquid

B. gas dispersed in liquid. D. solid dispersed in gas

C B 3. Which of the following statement is not true about oxidizing agent? reduced element → gain e⁻ undergoes reduction

A. A reducing agent donate electron in redox reaction ✗

B. An oxidizing agent gain electron in redox reaction

C. A reducing agent contains an element undergoes reduction

D. An oxidizing agent causes oxidation during redox reaction

B 4 .which of the following is an example of liquid solution?

A. Natural gas ✗ B. benzene in carbon tetrachloride C. Ceramic. D .steel.

C 5. All of the following are heterogeneous mixture except

A. Blood. B. Milk C . Air. D. .smoke

B 6. Which of the following is a characteristic of colloid?

A. it is a homogenous mixture. ✗ C. Particles settle on standing

B. Shows the Tyndall effect. D. Can separated by ordinary filtration

B 7. In which of the following colloid is the dispersing medium a liquid?

A. Mist. (gas) B .foam. C. Jelly. D. Shaving cream

D 8 . Which of the following is not an example of colloid?

A. Milk. B. Mud. C. Butter. D. Boric acid

C 9. Which of the following is colloidal system represent a gel?

A. Solid in liquid. B. Solid in gas. C. Liquid in solid. D. Liquid in gas.

AB 10. The empirical formula of a compound is CH_2 its molecular mass is 56 .

what is its molecular formula?

A. C_3H_6. B. C_4H_8. C. C_5H_{10}. D. C_2H_4

A 11. Which of the following substance is miscible within each other?

A. CH_3OH and CCl_4 C. C_2H_5OH and benzene

B. CH_3OH and H_2O D. hexane and water

A 12. Which of the following process is reduction reaction?

A. $Cu^{2+} + 2e^- \rightarrow Cu$ C. $Na \rightarrow Na^+ + 1e^-$

B. $2Cl^- \rightarrow Cl_2 + 2e^-$ D. $Fe \rightarrow Fe^{3+}$

A 13. Which of the following substance is polar?

A. NH_3 B. O_2 C. CH_4 D. H_2

II. Write true if the statement is correct and false if the statement is wrong.

false 14. The component of homogenous mixture is visible

false 15. Any mixture can be separated by filtration

True 16. The particle of solution less than the particle of colloid

True 17. Solution is a homogenous mixture

III. **Give short answer**

18. Calculate the percentage composition by mass of each element in $CaCO_3$?

(Mass; Ca=40, C=12, O=16)

% Calcium= 40%

% Carbon= 12%

% oxygen= 48%

BONUS

Balance the following chemical reaction using any method

A. $Mg + H_2O \rightarrow Mg(OH)_2 + H_2$

$Mg + 2H_2O \longrightarrow Mg(OH)_2 + H_2$

B. $CaCl_2 + Na_3PO_4 \rightarrow Ca_3(PO_4)_2 + NaCl$

$3CaCl_2 + 2Na_3PO_4 \longrightarrow Ca_3(PO_4)_2 + 6\,NaCl$